Priyanka Magar

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RESEARCH INTERESTS

My research interests include geometric topology, stable homotopy theory, and K-theory. My doctoral research focused on the classification of smooth structures on higher-dimensional manifolds, particularly their classification up to isotopy in dimensions 8 to 10.

ACADEMIC EXPERIENCE

Research Associate — 12 July, 2024 – Present
Indian Institute of Technology Madras

Research Associate — 28 May, 2024 – 11 July, 2024
Indian Institute of Technology Bombay

EDUCATION

Indian Institute of Technology Bombay, Mumbai, India — July 2018 — May 2024
Doctor of Philosophy — Defense date - 22 Nov 2024
Thesis: Smooth structures on PL-manifolds of dimensions between 8 and 10. — Cumulative GPA: 7.96

University of Pune, Pune, India — July 2015 — May 2017
Masters of Science in Mathematics — Cumulative GPA: 8.05

S. P. College (Aff. University of Pune), Pune, India — July 2013 — May 2015
Bachelor of Science in Mathematics — Percentage: 87.25%

PUBLICATIONS/PREPRINTS

- **P. Magar-Sawant**, *Concordance structure set of connected sum of projective spaces.* Proceedings-Mathematical Sciences, https://doi.org/10.1007/s12044-024-00800-7.
- S. Basu, R. Kasilingam, **P. Magar-Sawant**, *Smooth structures on PL-manifolds of dimensions between 8 and 10.* Positive review with Minor revisions. In: Proceedings of the Edinburgh Mathematical Society. https://arxiv.org/abs/2302.02301.
- **P. Magar-Sawant**, *Tangential smoothings of k-fold connected sum of complex projective spaces*, https://arxiv.org/abs/2306.16911, Submitted.

CONFERENCES/WORKSHOPS/TALKS

Oral Presentation

- "Concordance structures of sets of 8 to 10-dimensional manifolds", *Young Topologists Meet-2024 Münster,* Germany, August 2024.
- "Smooth structures on connected sum of projective spaces", *International Conference on Topology and its Application-2023 Nafpaktos,* Greece, July 2023.
- "On the stable cohomotopy and KO groups of connected sum of projective spaces", 37^{th} *Annual Conference of the Ramanujan Mathematical Society*, Chennai, December 2022.

Workshop

- Operads in Topology, NCM Workshop, IIT Bombay, India, Nov 2024.
- Hochschild Homology, NCM Workshop, IISER Kolkata, India, Dec 2023.
- Cohen Macaulay simplicial complexes in graph theory, NCM Workshop, Chennai Mathematical Institute, India, July 2023.
- Dualities in Topology and Algebra, International Centre for Theoretical Sciences, TIFR Bengaluru, May 2023.
- Dualities in Topology and Algebra(Online), International Centre for Theoretical Sciences, TIFR Bengaluru, Feb 2021.
- Global Homotopy Theory, Department of Mathematics, IIT Bombay, Feb 2020.
- Diamond Jubilee Symposium, Department of Mathematics, IIT Bombay, Jan 2019.
- International conference on Algebra and Analysis, Department of Mathematics, SPPU, Pune, Dec 2017.
- Annual Foundation School-II, IISER Pune, 2018.
- Annual Foundation School-I, Bhaskaracharya Pratishthana, Pune, 2017.

Technical Talks

- "Classifying Spaces of Categories", *Topology and Related Topics Seminar*, IIT Bombay, Feb 2024.
- "Quillen equivalent stable homotopy categories", *Topology and Related Topics Seminar*, IIT Bombay, August 2023.
- "A brief introduction to the stable homotopy category", *Topology and Related Topics Seminar*, IIT Bombay, August 2023.
- "Essential Categorical Terms for Stable Homotopy Theory", *Topology and Related Topics Seminar*, IIT Bombay, August 2023.
- "Introduction to Spectra and Stable Homotopy Category", *Dualities in Topology and Algebra 2023*, ICTS Bangalore, May 2023.
- "Spectra and stable homotopy category" in tt-category and chromatic homotopy theory online seminar series, March 2023.

Expository Talks

- "Simplicial sets arising from categories" at Student Seminar 37, September 2023, IIT Bombay.
- "A brief introduction to the fiber bundle" at Student Seminar 19, April 2023, IIT Bombay.

Teaching Assistantship

Institute

- Numerical Analysis – MA 214, Spring 2021–22
- Differential Equations – MA 207, Autumn 2021–22
- Linear Algebra – MA 106, Spring 2020–21

- Calculus II – MA 111, Spring 2020–21
- Differential Equations – MA 207, Autumn 2020–21
- Calculus I – MA 105, Autumn 2019–20

NPTEL

- Introduction to Algebraic Topology-Part II: Prof. Anant Shastri, Autumn 2021
- Linear Algebra: Dr. Dilip Patil. Autumn 2024.

AWARDS/SCHOLARSHIPS/FELLOWSHIPS

UGC NET. AIR 12
UGC JRF. AIR 78

SKILLS

- **Environment:** LaTeX, Microsoft Office
- **Operating System :** Windows, Linux
- **Computer Algebra Systems :** SageMath